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Information Provision Activity Requirement Extraction from EU-Acquis: Experiments and Findings



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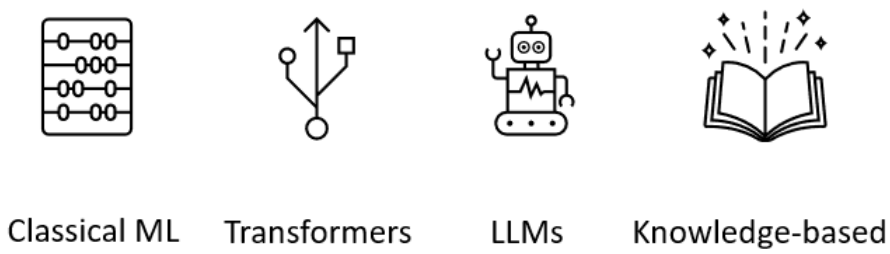
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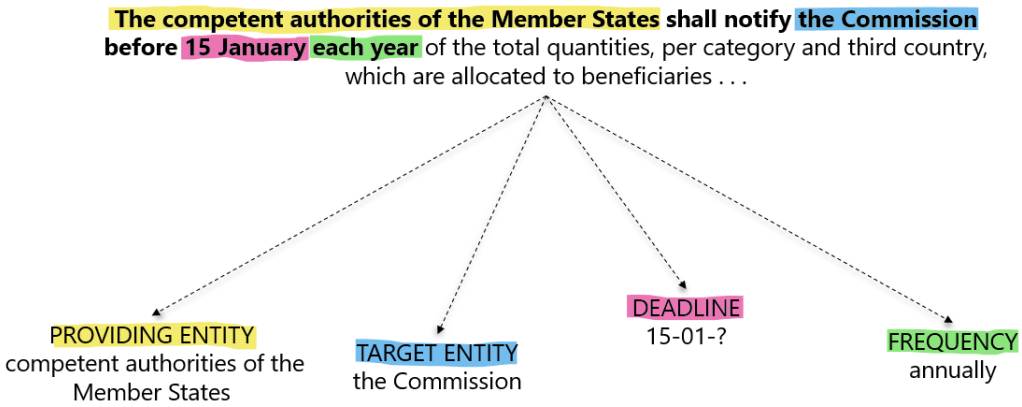
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Information Provision Activity Requirement Extraction from EU-Acquis: Experiments and Findings

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Abstract

In this report, we present the results of experiments centered on the extraction of structured information from legal acts in *EU Acquis*, the accumulated legislation of the EU. We introduce a new Information Extraction task of *Information Provision Activity Requirement Extraction*, which focuses on the identification in the text of legislation text fragments which do introduce an obligation or a possibility of information provision activities and extraction of structured key information therefrom, including providing and target entities, and respective deadlines and frequency of such provisions. We provide a comparative evaluation of various approaches for this task, including knowledge-, machine learning-, and generative AI-based ones, using a specifically designed benchmark corpus for this purpose.

The findings contained in this report constitute certain guidelines for elaborating, designing and eveloping technological solutions for extracting relevant structured information from EU Acquis and their industrialization.

1 Introduction

We are witnessing an ever-growing interest of legal professionals, government, and international organizations in exploiting Natural Language Processing-enabled tools to facilitate analysis and extraction of structured information from legal texts. While this interest has been reflected in the reported work of researchers for years (Ariai and Demartini, 2024), it predominantly focused on caselaw only, and hardly covered higher-level Information Extraction (IE) tasks such as relation extraction and template-filling extraction (Piskorski and Yangarber, 2013).

In this report, we present the results of the experiments revolving around extracting structured information from legal acts in *EU Acquis*, the accumulated European Union legislation[1]. More specifically, we introduce a new IE task of *Information Provision Activity Requirement Extraction* (IPARE), which focuses on the identification in the text of legislation text fragments which do introduce an obligation or a possibility of information provision activities and extracting structured key information therefrom, including providing and target entities, and respective deadlines and frequency of such provisions.

The motivation behind our work is three-fold. First, we are interested in exploring practical solutions for a real-world IE task in the legal domain. On the one hand, exploiting knowledge- and classical Machine learning-based approaches in industrial settings has proven to have numerous advantages in terms of time efficiency, fast development cycle and high level of explainability (Dahlmeier, 2017). On the other hand, the advent of generative AI opened new opportunities in terms of improved quality and ease of use. The question emerges what is the most appropriate technology to deploy for IE in the legal domain. Secondly, while a significant chunk of work in NLP focused on the caselaw, there is relatively little research reported on legislation, such as legal acts contained in *EU Acquis*, mainly covering text classification only, and nonexistent as regards IE. Finally, we intend to create and share linguistic resources to foster research and development in the area of IE in the legal domain, specifically for the subdomain of EU Acquis.

Our contribution can be summarized as follows:

— we introduce a new task on the extraction of structured information on information provision activity requirements from legal texts,

— we release a benchmark corpus for this task, focusing on *EU Acquis*,

— we report on comparative evaluation of various approaches for this task, including knowledge-, ML-, and generative AI-based ones.

It is important to emphasize at this stage that IPARE is a generic IE task and is meant to serve as an approximation of various more specific IE tasks such as extraction of, i.a., reporting requirements, different types of legal obligations, certification requests, data retention obligations, etc., from legal acts contained in the EU Acquis. Consequently, the findings and recommendations included in this report constitute certain guidelines for elaborating technological solutions for extracting relevant structured information from EU Acquis and their industrialization. We strongly believe that the material contained in this report will benefit Natural Language Processing (NLP) experts and software architects and engineers in the process of designing and developing real-world IE solutions for the legal domain.

The rest of this report is organized as follows. In Section 2 we briefly present related work. In Section 3 we introduce the information provision activity detection and extraction tasks. Next, in Section A we report on the creation of the benchmark dataset. Subsequently, in Section 5 we list the different approaches used in our comparative study. In Section 6 we present the evaluation results. We address the time complexity aspects in Section 7. We discuss the practical implications of the reported results in Section 8. We end with some conclusions and future work outlook in Section 9. Some supplementary material is provided in Appendices.

2 Related Work

Recently, one could observe a rapid growth of application of NLP techniques to the legal domain. In particular, a series of various workshops, e.g., *Natural Legal Language Processing* (NLLP) (Preoțiuc-Pietro et al., 2023,Aletras et al., 2024) and *Mining and Learning in the Legal Domain* (MLLD) (Makrehchi et al., 2023) have been organized and co-located with the major computational linguistics conferences. A survey on the advancements in Information Extraction for the legal domain has been presented in (Solihin et al., 2021).

Most prominent work in IE for the legal domain focused on Named Entity Recognition (NER), covering predominantly the subdomain of caselaw, contracts, and to a lesser extent the subdomain of national legislation. Various approaches to NER for the legal domain have been reported, including hand-crafted

[1] `https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=legissum:acquis`

rules (Glaser et al., 2018), classcial machine learning (Chalkidis et al., 2017), and transformer-based approaches (Cabrera-Diego and Gheewala, 2023). The *SemEval 2023* campaign included *LegalEval2023*, a shared task on NER for the legal domain (Modi et al., 2023).

Work on legal Relation Extraction (RE) is rather significantly less represented than NER. Similarly to NER, various NLP techniques have been explored for RE. (Andrew, 2018) introduced a French dataset for RE from legal documents related to investments in Luxemburg, and explored rule-based and ML approaches for this task. (Kwak et al., 2023) presents a dataset for RE from legal wills encompassing 18 relationship types and explores the performance of GPT 4o. (Thimm and Schneider, 2022) presents a rule-based NLP pipeline to extract relations from Environmental Law. (Kurant, 2023) explored the use of LSTMs for the task of extracting 'cause-and-effect' in Polish court judgements, whereas (Schraagen and Bex, 2019) compares LSTMs and SVM for the task of RE from Dutch police reports.

(Pennisi et al., 2023) presented work, on exploring classical and deep learning ML approaches for the task of detecting sentences with obligations in legislation texts originating from different countries (i.e., Code of Federal Regulations and Australasian Legal Information Institute). Work on detecting obligations in Italian legislation was presented in (Iacono et al., 2022), while (Minkova et al., 2023) reported on the classification of EU-Acquis documents into 5 categories, i.e., obligations, permissions, prohibitions, rights, and powers. Although all three aforementioned endeavours are partially similar to ours, they do not cover the IE aspect. As regards IE similar-in-nature efforts to ours on deploying classical and transformer-based ML approaches for extracting structured information on legal events (i.e., extracting event type, actors involved and the date of the event) from court judgements (including European Court of Justice rulings) have been reported in (Filtz et al., 2020) and (Navas-Loro and Rodríguez-Doncel, 2022). To the best of our knowledge, we are the first to report on exploration of IE focusing on information provision activities (e.g. reporting requirements) from EU-Acquis and exploiting LLMs for this purpose.

3 Task Definition

The task of *Information Provision Activity Requirement Extraction* (IPARE) focuses on identifying in the text of EU-Acquis text fragments (sentences or snippets) which do introduce an obligation or a possibility (based on some prerequisites) of information provision activities and extracting structured key information therefrom. Information provision activity is the process by which a *providing entity* communicates or provides information to a *target entity*. The relevant entities could be mentioned explicitly or implicitly. The providing entity is (or may be) obliged to perform an information providing activity, which involves systematic or one-time actions such as reporting, disclosing, notifying, certifying, or transferring information. Monitoring and publishing activities imply a transfer of information; hence, they are included as well. In the context of information provision activity requirement, we also distinguish between *primary* and *secondary* information, where the former includes information on the key actors involved, information/activity type, and activity-related deadlines and frequencies, whereas the latter encompasses details such as language and format of information to be provided, technicalities related to information process, other detailed descriptions of the information to be provided that go beyond primary information, etc. In IPARE task we focus solely on the extraction of primary information, i.e., detecting the text fragments that introduce the information provision activity requirements and contain all or some partial related primary information.

Formally, we distinguish two subtasks in IPARE. Given a text fragment T consisting of one or more sentences:

— Subtask 1 (detection): Determine whether T contains mention of an information provision activity requirement containing some primary information.

— Subtask 2 (extraction): If T contains mention of an information provision activity requirement with some primary information extract from T structured information mentioned in T on: (a) *providing entity*, (b) *target entitity/entities*, (c) any related *deadline(s)*, and (d) *frequency* of provision of information.

Subtask 1 is a binary classification task, whereas Subtask 2 is a template filling task. An example of a text snippet with mention of an information provision activity with key elements to extract is provided in Figure 1. A corresponding template with extracted information from text is provided below.

```
PROVIDING ENTITY: competent authorities of the Member States
TARGET ENTITY: the Commission
DEADLINE: 15-01-?
FREQUENCY: annually
```

. . . . . or any evidence considered by the aforementioned authorities to be equivalent thereto. 3. The competent authorities of the Member States shall notify the Commission before 15 January each year of the total quantities, per category and third country, which are allocated to beneficiaries . . . . . .

Figure 1: An example of a text snippet with mention of an information provision activity requirement with key elements to extract highlighted in text: entities (yellow), deadline (green), and frequency (blue).

Please note that in the case of deadlines and frequencies, whenever possible the slot values are to be normalized as part of the extraction process (see A.2 for further details on slot filling). Some more positive examples and the corresponding annotations are provided in Figure 2.

```
Such information shall be forwarded to the Bulgarian authorities by the end
of the period following the month to which the statistics relate.

PROVIDING ENTITY: -
TARGET ENTITY: the Bulgarian authorities
FREQUENCY: -
DEADLINE: -

-----------------------------------------------

Member States shall, if necessary, provide the Commission by 30 June every two
years with an update of the information described in paragraph 2.

PROVIDING ENTITY: Member States
TARGET ENTITY: the Commission
FREQUENCY: biannually
DEADLINE: 30-06-?

-----------------------------------------------

The feedback shall be reciprocal: the JST coordinators shall provide feedback
to the NCA sub-coordinators and the NCA sub-coordinators shall provide feedback
to the JST coordinators, in accordance with the principles set out in Annex I.

PROVIDING ENTITY: JST coordinators;NCA sub-coordinators
TARGET ENTITY: NCA sub-coordinators;JST coordinators
FREQUENCY: -
DEADLINE: -
```

Figure 2: Positive examples from the IPARE benchmark

The cases not considered to be in the scope of information provision activity requirement extraction (i.e., not to be detected as information provision activity requirement mention) are listed below:

— mentions of events such as announcements, publication and other actions resulting from information provision activity requirements (e.g., *By letter of 28 June 1996 the Member State submitted the requested reports to the European Commission*),

— mentions of only specific details (secondary information) that go beyond the provision of the primary information on who is required to provide what information to whom, by when and with what frequency, for instance, mention of a language in which the information is to be provided, e.g. *Notifications shall be submitted in one of the official languages of the Union.*,

— mentions of requirements not to disseminate or provide certain type of information (e.g., *Member States should not disclose the statistical data ...*),

— mentions of actions and events triggered by the obligation to provide certain information (that on their end do not cover transferring information or data), e.g., *Once the Member State submits the yearly report a workshop should be organized.*,

— mentions of actions and decisions to be taken if the information is not provided (e.g., *Where the Commission does not provide observations within that deadline, the reports shall be deemed to be accepted.*

Some additional negative examples of information provision activity requirement are provided in Figure 3.

```
The Commission decision to initiate the procedure was published in the Official
Journal of the European Communities(5).

-----------------------------------------------

ESMA shall determine whether the class of derivatives or relevant subset thereof
is only sufficiently liquid in transactions below a certain size.

-----------------------------------------------

If agreed in the framework contract, the payment service provider may charge
the payment service user for recovery.
```

Figure 3: Negative examples from the IPARE benchmark

4 Dataset

For the purpose of carrying out our experiments we have created a new benchmark datasaset since we were not aware of any resource with information provision activity requirement annotations. First, we have chunked the entire set of circa 110K documents in EU-Acquis into short text fragments (mainly sentences). Based on the intuitive assumption and some empirical observations that information provision activity requirements are expressed in EU-Acquis through the use of specific verbs, corresponding nominalizations and phrases (e.g., '*report*', '*submit*', '*transfer*', '*notify*', '*provide access*', '*reporting requirement*', etc.) we have randomly sampled from the pool of sentences: (a) 500 sentences which contain any of the predetermined keywords or any of their morphological variants or derived words (e.g., '*notification*'), and (b) another set of 500 sentences which apart from the keywords also contain at least one temporal expression, including dates, deadlines and frequencies. The sentences were selected in such a way that no two sentences are part of the same document in the EU-Acquis. After discarding some ill-formed texts and ignoring texts which were excessively long the final pool of sentences consists of 973 instances. The full set of keywords used for the preselection of the sentences is provided in A.1.

In order to make our dataset as representative as possible we have exploited a sample of the dataset presented in (Daudaravicius and Friis-Christensen, 2024) (which we will refer to with JRC-RO), which contains legislative documents from EU-Acquis which are in force and include mentions of reporting obligations, thus being in the scope of interest in the context of information provision activity requirement. In particular, all sentences that include information related to reporting obligations are annotated in the mentioned corpus separately, and no distinction is made whether they contain primary or secondary information (using the nomenclature of the IPARE task). We have extracted approx. 12K sentences from the aforementioned corpus **which do not contain any keywords from the set specified in A.1**. We further inspected a random sample of ca. 500 sentences therefrom and found out that only approx. 1% of them fall under information provision activity requirements, which, to some degree, confirmed our assumption of relatively low lexical variability in terms of key terms being used when referring to information provision activity requirements. Finally, 239 additional sentences from JRC-RO were added to our corpus, resulting in an overall number of 1212 instances in our dataset.

Two annotators were tasked to annotate each of the instances with a boolean label indicating the inclusion in the information provision activity requirement class, and they filled in the 4 slots for providing entity, target entity, deadline and frequency, using the annotation guidelines provided in A.2. In total for approx. 10% of instances there were some disagreements in labelling, which were resolved through some discussions.

The final dataset consists of 1212 items, out of which ca. 55% (662) are instances of information provision activity requirements. The fraction of these positive instances which contain providing entities, target entities, deadlines and frequencies amounts to 91.8%, 86.9%, 67.2% and 57.7% resp. Ca. 43.2% of all items contain both a keyword and either an explicit deadline or frequency mention. Analogously, this

figure for the positive instances only amounts to 79%. Approx. 19.6% of all instances do not contain any of the predetermined keywords.

Figure 4 shows that most sentences in IPARE are shorter than 500 characters. The average sentence length in the IPARE benchmark is 262 characters (or approx. 42 words). The longest sentence has 5252 characters and has been excluded from Figure 4.

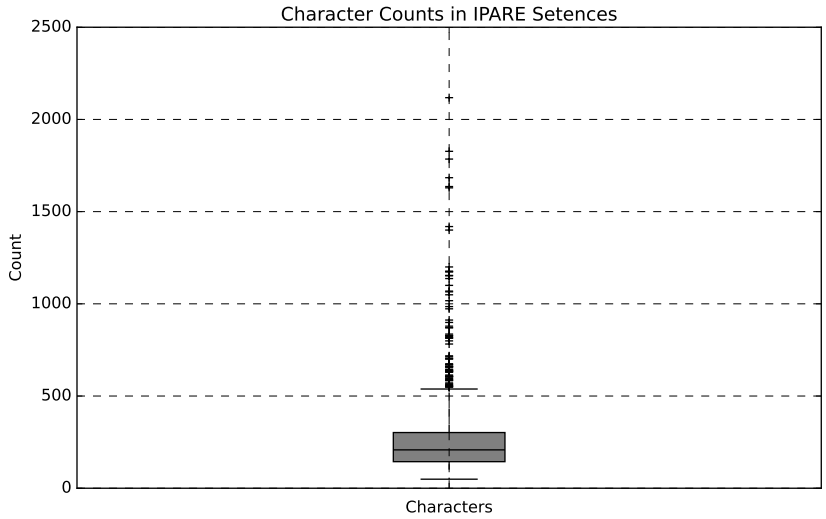


Figure 4: IPARE Sentence Length (in characters)

Figures 5 and 6 report on the cumulative distributions of providing and target entity mentions in the IPARE benchmark. Over 80% of providing entities are mentioned only once, and over 95% are mentioned at most 5 times. After accounting for small variations in mentions of those entities, at least 172 (out of 662) reporting entities are Member States. The Commission is required to provide information in at least 102 out of 662 sentences in the IPARE benchmark. As for the target entity, the cumulative histogram in 6 shows that over 90% of target entities are only mentioned once, and around 97% of them are mentioned at most 10 times. In fact, the Commission is the recipient of information in at least 283 out of 662 sentences in the IPARE benchmark. The European Parliament and the Council are also often mentioned together as the recipients of information.

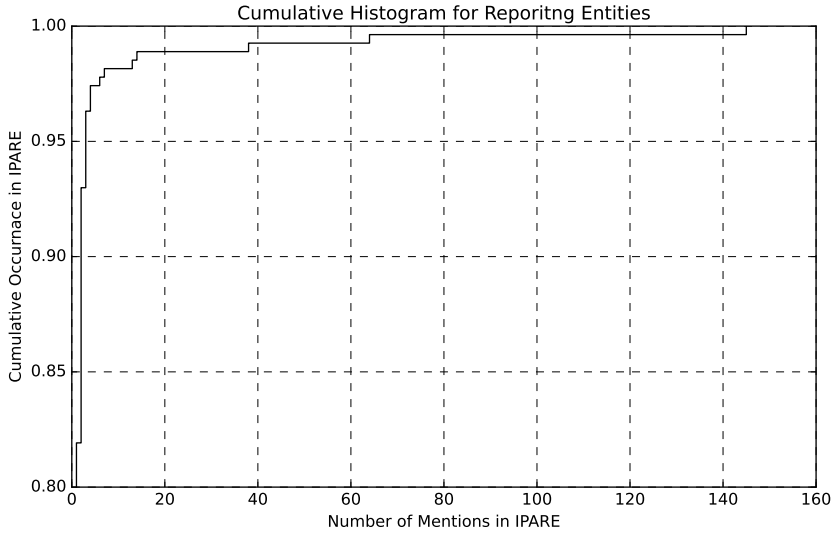


Figure 5: Cumulative Distribution of Reporting Entity Mentions in IPARE

5 Models

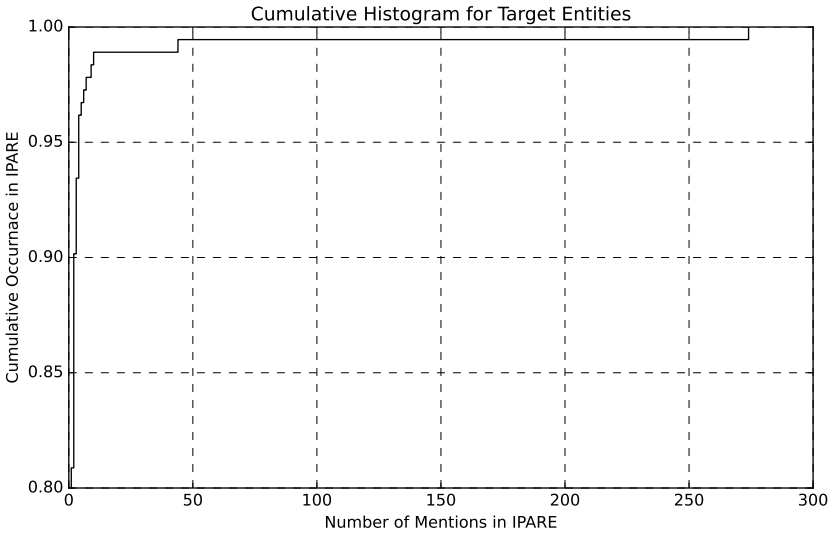


Figure 6: Cumulative Distribution of Target Entity Mentions in IPARE

5.1 Subtask 1: detection

To compare model performance on subtask 1, we divided the benchmark corpus into 5 folds. All models were trained/fine-tuned/instructed using 80% of the benchmark and were, then, evaluated on the remaining 20%, for each of the 5 folds. We have compared the following approaches for the task of detection of information provision activities,

BASELINE-K: considers a piece of text as a information provision activity if one of the keywords used to create the dataset appears in the text.

BASELINE-KD: considers a piece of text as an information provision activity if one of the keywords used to create the dataset appears in the text, and on top of this the text contains at least one temporal expression[2], e.g., date, duration, period, deadline, frequency, etc.

LR: L2-regularized Logistic Regression with binary 3-8 character n-grams (with minimum 3 occurrences) as features, using vector normalization and $c = 1.0$ resulting from parameter optimization.

SVM: L2-regularized linear SVM, with binary 3-10 character n-grams (with minimum 3 occurrences) as features, using vector normalization and $c = 1.0$ resulting from parameter optimization.

Contextual-based embedding based models (Transformers): including, **XLMRoBERTa** (Conneau et al., 2020), and two models tailored to the legal domain, namely, **Legal-RoBERTa** (Chalkidis et al., 2023) and **EU-BERT**[3]. A classification head was appended to each model, and the original layers had been frozen before fine-tuning for the detection of IPARE.

LLMs: including four models of different size, namely, **LLama-3.2-3B**[4] (Grattafiori et al., 2024), **Mistral-7B** (Jiang et al., 2023), **Mistral-24B**[5], **Mistral-Large (24.02)**[6] , **GPT 4o**[7] and **Claude-3.5-Sonnet**[8]. For subtask 1, we evaluated the four LLMs in zero-shot and few-shot (n=10) scenarios. For the latter, we instructed each model with examples sampled at random from the train folds. See Annex B for the hyperparemeters and detailed prompts.

5.2 Subtask 2: extraction

5.2.1 Knowledge-based Approach

The main idea behind our knowledge-based (**KB**) approach was to exploit certain level of lexical and syntactic repetitiveness exhibited in the EU-Acquis texts. In particular, based on word n-gram frequency analysis of the entire EU-Acquis corpus consisting of ca. 110K documents, (excluding the ca. 1K documents which were used for the creation of the evaluation dataset) we have created a pool of ca 2700

[2]We have developed a few regular patterns to capture temporal expressions
[3]Sébastien Campion, https://huggingface.co/EuropeanParliament/EUBERT
[4]https://huggingface.co/meta-llama/Llama-3.2-3B
[5]https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501
[6]https://mistral.ai/news/mistral-large
[7]https://cdn.openai.com/gpt-4o-system-card.pdf
[8]https://www.anthropic.com/news/claude-3-5-sonnet

```
(1) DEADLINE? FREQUENCY? PROVIDER <provide-to> TARGET+
(2) PROVIDER <reports-to> TARGET+ DEADLINE? FREQUENCY?
(3) DEADLINE? FREQUENCY? <reported-by> PROVIDER "to" TARGET
(4) PROVIDER <obligation> DEADLINE? FREQUENCY? <provide-to> TARGET
```

Figure 7: Examples of high-level information extraction patterns, where <provide-to>, <reports-to>, <reported-by> represent different classes of information provision trigger phrases, backed up by lexical resources, and <obligation> covers verb groups triggering an obligation.

labelled lexical resources which cover: trigger phrases for the recognition of information provision activity requirements (e.g., '*shall submit to*'), relevant entities or trigger words for detecting entities (e.g., '*competent authorities of the Member States*'), and keywords for the detection of relevant temporal expressions (e.g., '*annually*'). On top of the aforementioned resources a 3-level cascade of lexico-syntactic patterns was manually built. The first two levels of this cascade contain patterns for the detection of the small-scale structures like entities and temporal expressions, whereas the third level consists of ca. 45 high-level patterns which fuse the smaller structures together to fill in information provision activity requirement templates. Some examples of such high-level extraction patterns (in a simplified form) are shown in Figure 7. For instance, with pattern (4) one would cover the text fragment *The Member States shall by the end of March every year submit the report with aggregated data to DG HOME*.

Given that sentences in legal texts might be extremely long and complex, we have implemented a version of the **KB** approach which allows the extraction patterns to match longer sequences of tokens between the sought-after elements, hoping to boost the recall. We will refer to this approach with **KB+**.

For the encoding of the extraction patterns we exploited *ExPRESS*, a time-efficient finite-state based extraction pattern engine, whose rules are regular expressions over feature structures presented in (Piskorski, 2007).

5.2.2 LLM-based Approaches

We tested LLMs' ability to perform the task at hand when presented with the definition of IPARE along with a few examples of correctly extracted information. We created few-shot and Chain-of-Thought (CoT) prompts to evaluate the tested LLMs. We kept the same hyperparameters selected for subtask 1. All models were tasked to generate a JSON dictionary with the extracted text. For both prompting techniques, we used additional 13 annotated examples that were not included in the corpus. The CoT prompts consisted of an additional question for each of the slots to be filled before displaying the final JSON. See Annex B.3.1 for the detailed prompt templates.

Moreover, for **Claude-3.5-Sonnet**, we used its function calling ability to enforce the JSON format better and pass more detailed information about the format of each field. The model generated the input for the function calling tool, but no additional function was called. See Annex B.3 for the tool configuration.

Finally, assuming the **KB** was to achieve high precision figures we have also explored a hybrid approach, where first **KB** is applied, and whenever a slot could not be filled an LLM-based approach is triggered to try to fill in the value of that slot.

6 Evaluation

6.1 Subtask 1: detection

We evaluate the detection task using the standard *precision*, *recall*, F_1, and *accuracy* metrics. We provide the evaluation results in Table 1. The best observed results per group in terms of F_1 are highlighted in yellow.

We can observe that even the baseline methods achieve fair results with F_1 oscillating around .80, indicating the complexity of the detection task. Classical ML paradigms obtain surprisingly good results, e.g., **SVM** which exploits character n-gram as features obtains .927 F_1 score. The transformer-based models (**EU-BERT**, **Legal-RoBERTa**), including the ones specifically tuned to the legal domain did not perform better than the classical ML approaches, probably due to the small size of the training data. In particular, the best transformed-based model in terms of F_1 (.90) was **Legal-RoBERTa**. Similar observations were made in the context of somewhat related experiments on classification of legal texts in (Lin et al., 2023), were linear classifiers were shown to exhibit competitive or in some cases even better performance than more advanced models, which highlights the importance of inclusion of classical ML

Model	precision	recall	F_1	accuracy
Baseline-K	.671	.968	.799	.729
Baseline-KD	.873	.714	.786	.787
LG	.908	.919	.913	.906
SVM	.937	.917	.927	.920
BERT-derived				
EU-BERT	.881	.898	.889	.878
Legal-RoBERTa	.871	.924	.900	.882
XLM-RoBERTa	.821	.848	.833	.814
Zero-shot				
Claude-3.5-Sonnet	.889	.953	.920	.909
Mistral-Large	.793	.970	.872	.845
Mistral-24B	.949	.926	.937	.932
Mistral-7B	.811	.902	.854	.832
GPT-4o	.932	.935	.934	.927
Llama-3.2-3B	.966	.811	.882	.881
Few-shot (n=10)				
Claude-3.5-Sonnet	.951	.950	.949	.944
Mistral-Large	.941	.938	.939	.933
Mistral-24B	.949	.952	.950	.946
Mistral-7B	.918	.940	.927	.919
GPT-4o	.993	.855	.918	.918
Llama-3.2-3B	.850	.955	.899	.883

Table 1: Subtask 1: Detection - Evaluation results in terms of precision, recall, F_1 and accuracy. The best observed results per group in terms of F_1 are highlighted in yellow.

approaches such as logistic regression and **SVM** in comparative evaluations. In the zero-shot scenario, **Mistral-24B** and **GPT-4o** obtain the best F_1 scores, .937 and .934, respecitvely. The best overall results with LLMs were obtained in the few-shot scenario, with **Mistral-24B** and **Claude-3.5-Sonnet**, reaching .950 and .949 F_1, respectively. We also observe that extra examples (few-shot) nudged **GPT 4o** to be highly precise but decreased its overall performance. Figure 8 shows the results of **Claude-Sonnet-3.5** in the few-shot scenario for the detection task with different number of examples (selected randomly) used. One can observe that adding more examples to the prompt does not change the overall performance. When including between 10 and 100 examples in the few-shot prompt, the classification metrics oscillate around 0.95, as observed earlier for n=10. One can observe some gains in precision, and simultaneous loss in recall.

Although, the overall best result was obtained with an LLM in our comparison, the reported figures for the classical ML approaches show competitive performance and indicate that such methods constitute thanks to their favourable time complexity and robustness an interesting alternative. In particular, in our scenario of processing hundreds of thousands of legal texts, these classical ML paradigms are the most attractive and preferred option for the detection task, also considering that there is space to improve them, e.g., through the inclusion of more sophisticated features going beyond character n-grams.

6.2 Subtask 2: extraction

We evaluate the performance of the extraction of the different slots using *precision*, *recall* and F_1 measure. In particular, two types of evaluations were carried out:

— **Exact:** a system response is considered correct if it exactly corresponds to the annotation in the ground truth,

— **Fuzzy:** a system response is considered correct if the string similarity thereof with the ground truth is higher than .75[9], where for the measurement of the string similarity we used the *Longest Common Substrings* measure (Navarro, 2001),

[9]This treshold was set based on empirical observations

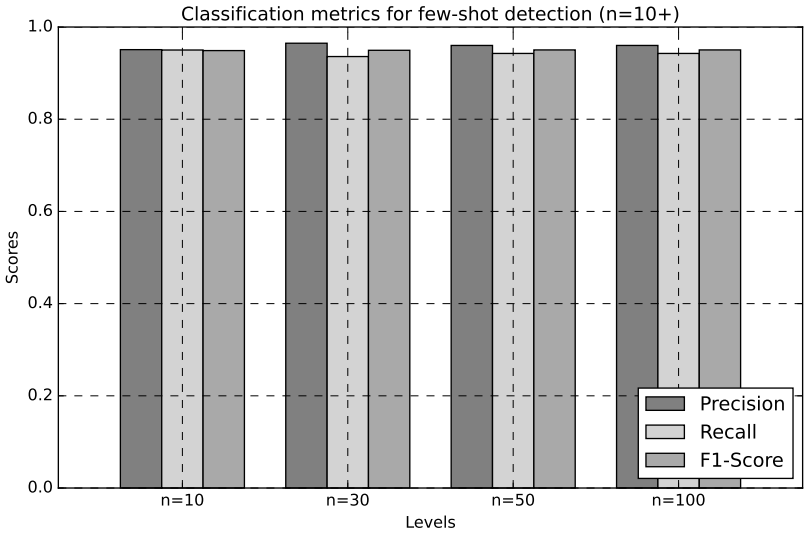


Figure 8: Classification metrics for few-shot LLM detection with n=10, 30, 50, 100 for Claude-Sonnet-3.5

Model	providing entity			target entity			deadline			frequency		
	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
KB	**.996**	.667	.806	**.998**	.814	.897	**.998**	.829	.906	**1.00**	.923	.960
KB+	**.996**	.697	.820	.995	.832	.906	**.998**	.853	.920	.998	**.949**	**.973**
						Few-shot (n=13)						
Claude-3.5-Sonnet	.919	.905	.912	.982	.939	.960	.922	**.942**	.932	.963	.912	.937
Mistral-Large	.967	.910	.937	.979	.941	.960	.923	.794	.854	.974	.873	.921
Mistral-24B	.993	.907	.948	.981	.938	.959	.942	.895	.918	.958	.886	.920
Mistral-7B	.917	.842	.878	.964	.841	.899	.733	.491	.588	.900	.839	.868
GPT-4o	.992	**.922**	**.956**	.984	**.956**	**.970**	.932	.844	.886	.977	.887	.930
Llama-3.2-3B	.934	.869	.900	.941	.892	.916	.772	.496	.604	.856	.766	.809
						Chain-of-Thought (with few-shots)						
Claude-3.5-Sonnet	.993	.909	.949	.989	.949	.967	.948	.904	.925	.979	.917	.947
Mistral-Large	.976	.890	.931	.982	.925	.953	.934	.673	.782	.987	.894	.938
Mistral-24B	.975	.903	.937	.985	.934	.959	.951	.921	**.935**	.977	.920	.948
Mistral-7B	.938	.877	.907	.965	.897	.930	.879	.649	.746	.927	.861	.893
GPT-4o	.992	.916	.953	.981	.943	.962	.931	.845	.886	.982	.916	.948
Llama-3.2-3B	.918	.854	.885	.984	.847	.910	.870	.607	.715	.722	.854	.782
						Hybrid						
KB + Claude-3.5-Sonnet	.991	.881	.933	.989	.924	.955	.954	.904	.928	.982	**.982**	**.982**

Table 2: Subtask 2: Extraction - Evaluation results per slot type in terms of precision (P), recall (R) and F_1 in exact matching for deadline and frequency slots, and fuzzy matching for providing entity and target entity slots. The best results are provided in bold.

We provide the evaluation results for the extraction of the various slots in Table 2. Given that the expected system responses (values) for the *deadline* ad *frequency* slots are to be normalized, they were only evaluated in terms of exact matching, whereas *providing entity* and *target entity* slots were evaluated in terms of fuzzy matching in order not to penalize inclusion of 1-2 extra tokens (e.g. articles, adjectives, etc.) in the response.

Overall, **Claude-Sonnet-3.5** achieved the highest level of performance on IPARE, i.e., the highest average F_1 over the four extraction slots. As for the individual slots, **GPT 4o** achieved the highest F_1 scores among the tested models when predicting the providing and target entities. **Mistral-24B** was the best (in terms of F_1) to obtain the provision activity deadline. **KB+** surpassed all the LLMs in the extraction of the frequency slot. The tool functionalities (only available for Claude) were effective in defining and enforcing a correct JSON schema. Other LLMs would benefit from a similar JSON tool when prompted to provide a structured output, such as extraction by template filling. Despite no extra tool used, **GPT 4o** and **Mistral-24B** were competitive to **Claude-Sonnet-3.5**.

We can observe that the knowledge-based approaches, **KB** and **KB+**, are highly competitive vis-a-vis LLMs, and at times outperform some smaller LLMs in terms of F_1, and, not surprisingly, outperform the best LLMs in terms of precision for the extraction of all four slots by .003, .009, .047 and .13 points, respectively for providing entity, target entity, deadline, and frequency. Both deadline and frequency slots required additional normalization. In fact, this could be the reason why the **KB** and **KB+** approaches were the best to fill the frequency slot, as highlighted by all three metrics. Apart from deadline extraction, we observed much lower recall compared to the other approaches. This makes LLMs the preferred approach for recall maximization, irrespective of the size of the model.Noteworthy, **KB+** improves the recall vis-a-vis **KB** by .014-.025 points.

The effects of our CoT prompt are apparent only for the deadline and frequency slots. By introducing an additional step in model reasoning, we were able to achieve better metrics for the two slots expecting an exact match. **Llama-3.2-3B** struggled the most with this strict format requirement. Apart from deadline extraction, applying the CoT prompt decreased this model's performance on IPARE. We hypothesize this could have been due to the size of the model. The few-shot prompt with the JSON tool helped **Claude-Sonnet-3.5** achieve better scores in deadline extraction without applying the CoT prompt. **Claude-Sonnet-3.5** benefited from CoT when extracting frequency (and the two entity slots). Since the deadlines and frequencies required applying more complex normalization rules, LLMs struggled the most with the extraction thereof. Function calling tools and targeting the temporal aspects of IPARE with additional questions could effectively guide LLMs in this task. At the same time, we did not find an overall trend that would show the effectiveness of our CoT prompt.

Given the emerging complementary nature of the tested approaches, we also investigated the scenario if the overall best-performing LLM were to be deployed after the **KB** algorithm in case the latter failed to extract value for some slots. Apart from the frequency slot, there were no improvements over the single-method approaches in terms of F_1, but, noteworthy, the obtained results improved in terms of precision, and the obtained F_1 results were not lagging much behind the respective LLMs, which make such a hybrid approach very attractive from the time efficiency point of view (see section 7 for more details on time complexity considerations).

Given the observable trend of the LLMs (which we delve into deeper in the subsequent section) to favour slightly "longer" answers on the participating entities and for the temporal slots we expected the LLMs to achieve better results in the fuzzy evaluation, which is actually confirmed by the results reported in Table 2.

6.2.1 Error Analysis

6.2.1.1 Detection

BERT-derived models misclassified between 11.8 and 18.6% of the instances, and approx. 30% of the misclassified instances were shared across the 3 tested models. Among the common false negatives, the majority of the requirements were presented in the passive voice. Some of the undetected information provision activity requirements were mentioned in the sentences in form of nominalizations, e.g. '*the subject of the reporting obligations*'. False positives were mainly sentences reporting facts or actions triggered by information provision activity requirements, which per se are out of scope.

In addition, two LLMs have been selected for error analysis, i.e., **Claude-Sonnet-3.5** and **Mistral Large**. When prompted with a few examples (few-shot), they produced 68 and 81 incorrect responses, respectively, and shared 43 instances with errors. Most common false negatives featured application, publication, and certification activities, whereas most represented group among false positives constituted instances with complex syntactic structure and ones with entities scattered across the sentences. The set of examples in the prompt might not have been fully representative, as only 10 examples were selected

at random from the training portion of the split. However, the experiments with an increased number of examples in the few-shot scenario did not boost the performance (see 6.1).

6.2.1.2 Extraction

The same two models, i.e., **Mistral-Large** and **Claude-Sonnet-3.5**, were inspected for the extraction subtask. In the providing entity slot, **Mistral-Large** made most of the errors by extracting too long text fragments. For instance, instead of extracting only the entity, an adjacent reference to a regulation was included in the system response (e.g. *'The authorities referred to in points (a) to (e) of Article 55(4) of Regulation (EU) No 909/2014'*. Other typical errors were the omissions of entities in longer syntactically complex sentences. We also observed some errors (<1%) of not capturing entities from sentences that introduce information provision activities using passive voice constructions.

As regards extracting the target entity, **Claude-Sonnet-3.5** and **Mistral-Large** showed a competitive level of recall and precision compared to the other models. In most cases with multiple target entities (e.g., *the European Parliament, the Council, and the Commission*), the LLMs did not respect the rules for extracting continuous text without breaking it into a sequence of target entities, however, this somewhat undesired effect was not penalized by the fuzzy evaluation metric.

As for **KB** approach, the majority of the errors (185 instances) related to providing entity were a result of not detecting an entity due to difficulty of capturing long-distance dependencies in longer sentences. Other errors were the result of extracting incomplete entities, e.g., *entity* instead of *The contracting authority or the contracting entity*. We also observed issues in sentences were entities were separated by a comma, but the second entity was the beginning of another clause (not a second entity). For target entity, the knowledge-based approaches made most errors by not detecting an entity (87 sentences), however, this type of error was less prominent vis-a-vis ommissions of providing entities.

Claude-Sonnet-3.5 achieved one of the highest F_1 scores (for LLMs) on frequency extraction with both prompting techniques, falling .01 short of the scores achieved by **GPT-4o** and **Mistral-24B** (see Table 2). Nearly half of the incorrect extractions (37 out of 78) were expressions referring to *biannual* and *semiannual* frequencies, which the model failed to normalize. **Claude-Sonnet-3.5** also assumed periodicity when exposed to text fragments such as *shall monitor*, *shall keep informed*, etc. In other words, it made associations related to frequency that were actually not present in the text. This type of error was not observed for **KB** and **KB+**. The **KB** and **KB+** algorithms were also very precise with the deadline extraction, missing only a few deadlines not expressed as a date (e.g., *within six months*, *at least 6 months before*). **Claude-Sonnet-3.5** achieved a good balance between high precision and high recall with a slightly better F_1 score with the few-shot prompt. We reason this was likely thanks to the function calling tool.

While LLMs perform well on IPARE, we infer from the error analysis that they struggle with understanding the syntactical structure of the sentence, which still hinders them to extract some relevant elements correctly. Meanwhile, the patterns used by **KB** can detect the role of the entities (target/reporting) much more precisely due to the more local nature of these patterns and linguistic awareness encoded in them.

7 Practical Considerations of Time Complexity

The models explored in our study differ not only in size, but also in terms of time complexity. We distinguish here between the time needed for creating (training) the models and the inference time.

For the smaller models, i.e., baselines, knowledge-based, and classical ML ones, the training time is low and does not require specialist hardware from data centers. As regards the knowledge-based models their creation time might increase with the granularity of the information to be extracted, which boils down to the creation of relevant linguistic resources (e.g., lexica and extraction patterns), nevertheless, the time needed should not be considered as a time-critical factor. Moreover, the time it takes to convert an IPARE sentence (input) into the indicator (subtask 1) or a filled extraction template (subtask 2) could also be viewed as negligible in the context of the aforementioned approaches.

As for the tested LLMs, not all authors shared exhaustive information on their model training. For the smallest tested LLM **LLama-3.2-3B**, the training time was 460k GPU hours, as per its model card [10]. For some tasks, LLMs can be used off the shelf if the prompt alone can guide the model effectively enough to succeed. In that sense, we did not adjust the LLM model weights for IPARE and only used a handful of annotated examples. In this scenario, the LLM inference time is of greater importance.

LLM inference time is influenced by a variety of factors. Some of them are beyond the control of the model user. Those are network conditions, geographic location, resource availability, etc. However,

[10] `https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD.md`

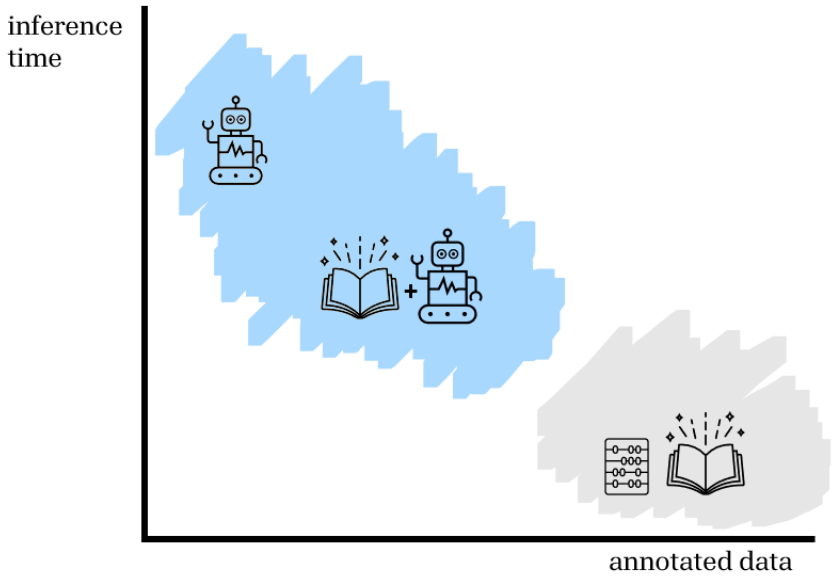


Figure 9: Trade-off between inference time and amount of annotated data required for operationalization of IPARE with lower annotation needs, high inference time approaches (in blue) and higher annotation needs, lower inference time approaches (in gray)

the user (partially) controls other factors, such as, model size, prompt lengths and complexities, post-processing, requested output lengths, etc. In the context of IPARE, subtask 1 generally requires less time due to shorter prompts and requested outputs. We found that the largest model (Claude-3.5-Sonnet) needed around 1-2 seconds for detection and 4-6 seconds for extraction. This varied depending on the input sentence length. The total time needed to complete an iteration of the entire benchmark was 25-30 minutes for detection (1212 sentences) and 35-45 min for extraction (662 positive sentences). Assuming an average document length of 50 sentences, the estimated time needed to execute subtask 1 (Detection) on a legislative act by one of the largest models (i.e., **Claude-Sonnet-3.5**) is about 73 seconds using a cloud workspace designated by the cloud provider. This approximation can be extended to the whole *Acquis*. Assuming there are approx. 24.5k acts in the EU *Acquis* corpus we would like to process, the estimated processing time (with **Claude-Sonnet-3.5**) should be at least 20 days. The time to process subtask 2 (extraction) would depend on the number of positive sentences in subtask 1. These estimates exclude possible fluctuations in latency and any computional/token limits imposed by cloud providers.

In a deployment scenario, one should consider the trade-off between the inference time and amount of annotated data required to make a model operational. The blue area in figure 9 shows the technologies that can be made operational with less annotated data (i.e., LLMs and a combination of LLMs and knowledge-based approaches). The approaches marked in gray have lower inference time, but they require more data for full operationalization (e.g., transformer- and other classical neural network-based models). Despite their different data needs, some annotation is crucial to delineate precise and exhaustive definitions of the task. That said, LLMs can rely more on task definitions expressed in natural language. This advantage comes at a considerably higher inference time. It is especially relevant for real-time processing, where latency can have a big impact on the business value of a model. Thus, LLMs could be combined with other methods that will pre-select a limited number of examples. They could also be used entirely offline to support annotation by generating synthetic labels for a human-defined task (Pavlovic and Poesio, 2024).

8 Discussion

8.1 Detection

Classical ML, such as SVMs, may be the preferred solution for subtask 1, despite not achieving the highest performance on IPARE. The magnitude of the performance gap to the best-performing solution (i.e., LLMs) is relatively narrow, but the infrastructural resources that are required to detect information provision activity requirements in *EU-Acquis* sentences are incomparably higher in the context of LLM-based solutions, considering in particular the length of the legislation document. Contrary to LLMs (see section 7), SVMs are a computationally light-weight solution that can achieve good results with a moderate amount of annotated data, provided that it is representative of the task. It strikes a good

balance between performance, speed, and resource intensity. Furthermore, even for such classical ML approaches there is space for improvement given that we report on exploiting relatively primitive type of features, namely, character n-grams.

8.2 Extraction

Although LLMs achieved the overall best result (excelling in particular in terms of recall) on the IPARE benchmark, a hybrid solution combining knowledge- and LLM-based approaches may be more desirable for real-world scenario application, given the advantages of the **KB** approach in terms of time efficiency. **KB** may be a more suitable solution for IPARE when precision is favored over recall. Ideally, there should be a choice between the two approaches at inference time. Furthermore, the results of the error analysis also hints at the necessity to use a knowledge-based solution for the extraction of the temporal slots whose expected values have to be normalized, or use of an additional knowledge-based postprocessing component to facilitate normalization of temporal expressions found in text. Whether elaboration of a more sophisticated prompting strategy for LLMs could alleviate the issue of proper normalization of temporal expressions and better understanding of the link between syntactic structure and entity role assignment is yet to be explored.

9 Conclusions

We reported on experiments on extracting structured information from text fragments contained im *EU Acquis*, the European Union law, on Information Provision Activity Requirements (IPARE), which encompasses the extraction of involved actors and temporal aspects related to such activities, e.g. deadlines and frequencsies. We carried out a comparative evaluation of knowledge-, classical ML-, transformer-, and generative AI-based approaches for this task on a new benchmark corpus specifically created for this study. While our study is by far not exhaustive, it provides a good approximation of the capabilities of the different approaches and suitability for deployment in an industrial setting for the task at hand. We also believe the reported research and findings constitute useful compendium of knowledge for NLP practitioners building analytical applications for the legal domain.

There are various directions in which the presented work could be extended in the future, including, i.a.:

— creating a larger benchmark corpus,

— exploring whether inclusion of context in which the sentences with key information appear might contribute to the improvement of the extraction quality,

— defining the IPARE task at document level instead of sentence level as done in our study,

— exploring additional family of ML models for the task, e.g., transformer-based models for the extraction subtask, and

— extending the task to the extraction of additional aspects of information provision activity requirements, e.g., type of information to be provided (e.g., report, study, assessment, etc.), related modalities (language, format), distinguishing the type of the information provision, e.g., reporting, certification, notification, authorisation, data retention, monitoring, record keeping, compliance, etc.

In particular, it will be the specific business needs and the yet-to-be-defined granularity of the sought-after information that will drive further work presented in our study.

Limitations

Dataset Representativeness

The selection of the documents and text snippets for the creation of the benchmark dataset exploited in our study was done solely using EU Acquis, the EU legislation. Hence it might be considered as representative for this specific subdomain of legal texts only. It is of paramount importance to emphasize though that this dataset should not be considered as representative in terms of any type of information provision obligation, specific organization or geopolitical entity, nor should it be seen as perfectly balanced in any particular way.

Biases

Although a very detailed annotation guidelines were used for the sake of creating the benchmark corpus and multiple rounds of curating the annotations were carried out, we are aware that some degree of intrinsic subjectivity might be present in this dataset.

Models

While our study covers a wide range of approaches for the task at hand, namely, knowledge-, classical and transformer machine learning-, and generative AI-based approaches it should not be considered as exhaustive, but rather as an approximation. We did not explore certain models because the size of required annotated data might have been prohibitive. For instance, classical and transformer machine learning-based models are known to obtain superior performance in IE tasks in a fully supervised (full-shot) scenario (Keraghel et al., 2024, Wang et al., 2025) with training data of appropriate size. Furthermore, the error analysis is only provided for a subset of all models explored for the detection and extraction tasks.

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Appendices

A Dataset Creation

This section provides some more details related to the creation of the dataset, including: keywords used to collect documents, annotation guidelines, examples, and some corpus statistics.

A.1 Keywords

For selecting candidate text fragments for the inclusion in the evaluation dataset we have used the following keywords/keyphrases and their respective morphological variants and derived forms: *report, transmit, provide, communicate, notify, passed on, inform, share information, share data, submit, publish, send, update, keep a register, monitor, present, send copy, forward, grant access, make available, collect information, collect data*.

A.2 Annotation Guidelines

The annotation of the textual data consist of: (a) deciding whether a given text fragment embraces a mention of reporting requirements, and if affirmative, (b) filling in the four slots related to *primary information*: *reporting entity*, *target entity*, *deadline*, and *frequency* using the following rules:

— **General Rules:**

- for filling in the slots (unless specified differently for specific slots below) one should exclusively use the strings found in the text fragments,
- if specific type of information is not included in the text fragment a hyphen should be used to indicate this fact in the respective slot,
- in case of any doubts whether the information contained in the text fragment should be used as a slot filler or not a *conservative approach* should be applied, i.e., no annotation is to be made in case of uncertainty,

— **Entities**

- in case the strings that are related to filling in a given slot (for related entities) do not constitute a consecutive sequence of characters (e.g., more than one mention of relevant entities that are spread over the text fragment), one should simply cut and paste the relevant strings and delimit them by a semicolon, e.g., from the text fragment '*The operator or aircraft operator*' the entity slot should be filled in with '*The operator; aircraft operator*' as a value,
- both named (e.g., '*European Commission*') and nominal mentions (e.g., '*competent fiscal authorities*') of entities are considered as candidate slot fillers,
- the articles, whether definite or indefinite, constitute part of the string to be used as a slot filler for entities (e.g. *The European Commission*),
- in case pronominal references to entities (e.g., '*they*') are used, and there is no corresponding named mention of such an entity, then the pronominal mention should be used as a slot filler,
- one should annotate 'minimal' mentions of entities and disregard any relative clauses that provide further details of the entities, e.g., in the text fragment '*The custom authorities, which were tasked with carrying out the exercise ....*' only '*The custom authorities*' should be considered for filling in the entity-related slot.

— **Dates**

- in case of precise date references only the date information in a canonical form `DD-MM-YYYY` format should be provided, e.g., for *1 June 2023* the corresponding slot value should be 01-06-2022+ (in case of incomplete date references, i.e., with missing day/month/year a "?" should be used accordingly, e.g. *June 2023* corresponds to ?-06-2023)
- phrases triggering deadlines that precede or follow concrete dates are not to be annotated, e.g. *not later than*, *before*, etc.,
- analogously, for deadlines using a duration a canonical form should be used as well using the following format `DD-MM-YYYY:DD-MM-YYYY` format, e.g., for the text fragment '*from 1 to 10 January 2023*' the corresponding slot filler would be 01-01-2023:10-01-2023,

- in case of strictly relative (e.g., '*next year*' or imprecise temporal expression (e.g., '*without any delay*', etc.) used in the text, the surface form in the text constitutes the respective slot value,
- references to specific hours are not to be annotated as part of the date annotations,
- references to dates of specific periods in time from which information is to be collected and provided/reported are not considered as dates that need to be annotated since they do not refer to the deadline of reporting or provision of information/data, e.g., for the text fragment '*The required data for March to April in 2023 is to be provided not later then by January 2024*' the deadline slot is to be filled in with ?-01-2024.
- in case of relative or imprecise temporal references that can not be anchored on a time scale only a minimal 'phrase' that conveys the main time constraint and discards potential event dependencies embraced in text should be used as a slot filler, e.g., from the text fragment '*The report has to be submitted as soon as possible after the approval of the Committee and receiption of the authorisation from ....*' only the phrase '*as soon as possible*' should be used as a slot filler.
- in case the text fragment contains references to more than one deadline, e.g., reporting is to be carried out twice a year with two different dates, then the corresponding slot should contain both dates, delimitated by a semicolon.

— **Frequencies**

- for filling in the *frequency* slot one should use one of the following values: *daily*, *weekly*, *biweekly monthly*, *bimonthly*, *quarterly*, *semiannually*, *annually*, *biannually*, *every X years*, *every X months*, *every X days*, *periodically* (in cases it is not clear what the actual period is, e.g., *at regular intervals*), provided that the information contained in the text corresponds to any of the aforementioned values,
- if the frequency information can not be expressed using any of the values listed before, the relevant string from the text fragment should be used instead as a value,
- analogously to dates, minimal 'phrases' that embrace the key information on frequency should be picked up as slot fillers

B Models details

This section provides some more details on the models used in our experiments, in particular, details ralated to their tuning, etc.

B.1 Classical ML approaches

We used the original Liblinear library[11] to run the experiments with the LR and SVM.

B.2 BERT-dervied models

The three models employ the RoBERTa architecture proposed by (Liu et al., 2019), but they differ in size and training data. **XLMRoBERTa** and **Legal-RoBERTa** use 16 attention heads and 24 hidden layers to generate 1024-dimensional embeddings, whereas **EU-BERT** has only 12 attention heads and 6 hidden layers that produce 768-dimensional embeddings. **XLMRoBERTa** was pre-trained on 2.5TB of filtered, multiligual CommonCrawl (Conneau et al., 2020). **Legal-Roberta** continued training the original RoBERTa model on the LeXFiles corpus. The training data comprised of approx. 6 million legal English documents including 11 sub-corpora that cover legislation and case law from 6 primarily English-speaking legal systems (EU, CoE, Canada, US, UK, India). Finally, **EU-BERT** was pre-trained on a corpus[12] of documents from the European Publications Office consisting of nearly 4 million documents in 39 languages.

For each model, we optimized learning rate (ranging between 7e-5 and 5e-3), batch size (ranging between 32 and 128) and training epochs (ranging between 1 and 10). Then, the most accurate model was loaded from a checkpoint for assessment for each fold.

[11] `https://www.csie.ntu.edu.tw/~cjlin/liblinear/`

[12] Sébastien Campion, https://huggingface.co/datasets/EuropeanParliament/Eurovoc

B.3 Large Language Models

While **Claude-3.5-Sonnet** and **Mistral-Large (24.02)** are closed-source models, we can reason about their large size by comparing them to the other flagship LLMs with more detailed information. **Mistral-Large (24.02)** was created to compete with older versions of the largest Claude and Llama models. Shortly after that, the version of **Mistral-Large** from July 2024 was released as open source and included 133B parameters. **Mistral-Large (24.07)**[13] competed with **Claude-3.5-Sonnet** and **Llama-3.1-70B**. We can deduce **Claude-3.5-Sonnet** and **Mistral-Large (24.02)** are large LLMs, presumably with more than 70B parameters.

As observed in (Renze, 2024), temperature had no significant effect on model accuracy when applied to MCQA problems, but lower temperature maximized reproducibility. For all the tested LLM scenarios, we set the temperature hyperparameter to 0 (at top_p=0.95) to achieve a high level of determinism. The prompt comprised of the task definition along with a more detailed description of the information provision requirement.

B.3.1 Prompts

In Figure 10 and 11 we provide the prompts used for the detection subtask in the zero- and few-shot scenario resp. In Figure 12 and 13 we provide the prompts used for the extraction subtask in the single-instruction few-shot and chain-of-thought scenarios resp. Finally, Figure 14 shows the JSON Tool template passed with the prompts for **Claude-Sonnet-3.5** in few-shot scenario.

```
Assess if the example includes a reporting obligation,given the task description.
Respond with '1' if there is a reporting obligation or '0' if there is no reporting
obligation. Do not justify your answer.

Task Description:  The task is to identify and extract primary information from
EU legal texts that specify reporting obligations. This includes determining whether
a text imposes or suggests a reporting duty, identifying the key actors, the type
of information or activity reported, and any related deadlines or frequencies.
The focus is on examples that mandate or enable entities to report, disclose, notify,
provide, certify, or transfer information systematically or as a one-time action.
Monitoring and publishing are considered reporting activities. Secondary information
like language, format, or technical process details are not the focus of this task.
Those examples including only secondary information should be classified as '0'.

Example: {sentence}
```

Figure 10: Prompt template for zero-shot detection

[13] https://mistral.ai/news/mistral-large-2407

```
Assess if the example includes a reporting obligation,given the task description.
Respond with '1' if there is a reporting obligation or '0' if there is no reporting
obligation. Do not justify your answer.

Task Description: The task is to identify and extract primary information from
EU legal texts that specify reporting obligations. This includes determining whether
a text imposes or suggests a reporting duty, identifying the key actors, the type
of information or activity reported, and any related deadlines or frequencies.
The focus is on examples that mandate or enable entities to report, disclose, notify,
provide, certify, or transfer information systematically or as a one-time action.
Monitoring and publishing are considered reporting activities. Secondary information
like language, format, or technical process details are not the focus of this task.
Those examples including only secondary information should be classified as '0'.

Here are a few examples of how to respond in a standard interaction:
<example> <sentence> These are the technical provisions for general liability
insurance and proportional reinsurance, without risk margin after deduction
of the amounts recoverable from reinsurance contracts and SPVs,
with a floor equal to zero.</sentence>
Assistant: <response> 0 </response> </example>

[...]

<example> <sentence> [...] </sentence>
Assistant: <response> [...] </response> </example>

Here is the sentance: <sentence>{sentence}</sentence>

Put your response in <response></responses> Assistant: <response>
```

Figure 11: Prompt template for few-shot detection

```
Extract primary information from the sentence, given the task description.
Format your response as a json.  Do not justify your answer. Do not alter the
original phrasing of the sentence.

Task Description: If the sentence mentions an information provision
activity requirement with some primary information extract from the sentence:
"reporting entity", "target entitity", "frequency", and "deadline" of provision of
information. The focus is on examples that mandate or enable entities to report,
disclose, notify, provide, certify, or transfer information systematically or
as a one-time action. Monitoring and publishing are considered information provision
activities. Secondary information like language, format, or technical process
details are not the focus of this task.


Here are a few examples of how to respond in a standard interaction:
<example> <sentence> Every three years, and for the first time on 5 June 2004,
 the Commission shall publish a summary based on the reports
referred to in paragraph 2.. . </sentence>

Assistant: <json> {'reporting_entity': 'the Commission', 'target_entity': '-',
'frequency': 'every three years', 'deadline': '05-06-2004'} </json> </example>

[...]

<example> <sentence> [...] </sentence>
Assistant: <json> [...] </json> </example>

Here is the sentence: <sentence>{sentence}</sentence>

Extract "reporting entity", "target entitity", "frequency", and "deadline"
from the sentence. Format your response as json. Multiple answers should be seperated
by a ';' if they are not mentioned consecutively. If the extrated entity is not
present, insert '-'. Put your response in a single <json></json>.
```

Figure 12: Prompt template for few-shot extraction

```
Extract primary information from the sentence, given the task description.
Format your response as a json.  Do not justify your answer. Do not alter the
original phrasing of the sentence.

Task Description: If the sentence mentions an information provision
activity requirement with some primary information extract from the sentence:
"reporting entity", "target entitity", "frequency", and "deadline" of provision of
information. The focus is on examples that mandate or enable entities to report,
disclose, notify, provide, certify, or transfer information systematically or
as a one-time action. Monitoring and publishing are considered information provision
activities. Secondary information like language, format, or technical process
details are not the focus of this task.

Answer the 4 questions below and extract "reporting entity", "target entitity",
"frequency", and "deadline" from the sentence. Multiple answers should seperated
by a ';' if they are not mentioned consecutively. If the extrated entity is not
present, insert '-'. Do not justify your final answer. Do not alter the original
phrasing of the sentence. Put your response in a single <json></json>.

1. Does the sentence specify who provides primary information? If so, who is it?
2. Does the sentence specify who receives primary information? If so, who is it?
3. Does the sentence specify the frequency at which primary information
   is provisoned? If so, which one of the following values is it,  daily, weekly,
   biweekly, monthly, bimonthly, quarterly,  semiannually, annually, biannually,
   every X years, every X months, every X days, periodically?
4. Does the sentence specify the deadline by which primary information is provisoned?
   If so, what is it?

Here are a few examples of how to respond in a standard interaction:
<example> Here is the sentence: <sentence> Every three years, and for the first time
on 5 June 2004, the Commission shall publish a summary based on the reports
referred to in paragraph 2.. . </sentence>
Assistant: 1. the Commission
2. -
3. every three years
4. 05-06-2004
<json> {'reporting_entity': 'the Commission', 'target_entity': '-',
'frequency': 'every three years', 'deadline': '05-06-2004'} </json> </example>

[...]

<example> <sentence> [...] </sentence>
Assistant: <json> [...] </json> </example>

Here is the sentence: <sentence>{sentence}</sentence>
```

Figure 13: Prompt template for Chain-of-Thought extraction

```
{
    "toolSpec": {
        "name": "extract_entities",
        "description": "Extract entities",
        "inputSchema": {
            "json": {
                "type": "object",
                "properties": {
                    "reporting_entity": {
                        "type": "string",
                        "description": "entity reporting, disclosing, notifing,
                        providing, certifing, or transferring, monitoring
                        information systematically or as a one-time action"
                    },
                    "target_entity": {
                        "type": "string",
                        "description": "entity receving information
                        systematically or as a one-time action"
                    },
                    "frequency": {
                        "type": "string",
                        "description": "the frequency slot one should use one of
                        the following values: daily, weekly,
                        biweekly, monthly, bimonthly,
                        quarterly,  semiannually, annually, biannually,
                        every X years, every X months,
                        every X days, periodically, -",
                    },
                    "deadline": {
                        "type": "string",
                        "description": "in case of precise date references only
                        the date information in a canonical form
                        verb and DD-MM-YYYY, format should be provided, e.g.,
                        for 1 June 2023 the corresponding slot value should
                        be 01-06-2022, (in case of incomplete date
                        references, i.e., with missing day/month/year a '?'
                        should be used accordingly, e.g. June 2023
                        corresponds to ?-06-2023. In case of strictly
                        relative (e.g., 'next year' or imprecise temporal
                        expression (e.g., 'without any delay', etc.) used in
                        the text, the surface form in the text constitutes
                        the respective slot value."
                    }
                },
                "required" : ["reporting_entity",
                              "target_entity",
                              "frequency",
                              "deadline"
                              ]
            }
        }
    }
}
```

Figure 14: Tool template for Claude-Sonnet-3.5 This JSON tool was passed along with the prompt in the few-shot extraction.

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